

Problem Set for 08 May 2025: A New Jordan form over $\mathbb{C}$

Definition (Eigen-). Let $A \in \mathbb{C}^{n \times n}$. We say that (v, λ) is a eigenpair of A provided $A\bar{v} = \lambda v$ for some $\lambda \in \mathbb{C}$. The vector v is referred to as a eigenvector and λ a eigenvalue, correspondingly. For matrices, we say A is csimilar to B provided $A = P B \bar{P}^{-1}$ for some non-singular matrix P .

Exercise 1 Characterise the eigenspace corresponding to the eigenvalue 0.

就是 $\overline{\ker A}$.

Exercise 2 Demonstrate that if λ is a eigenvalue, then so too is $e^{i\theta} \cdot \lambda$. Furthermore, show that for any eigenvector v , there exists a eigenvector associated with a eigenvalue ≥ 0 .

若有 $A\bar{v} = \lambda v$, 则对任意 $t \in \mathbb{R}$ 总有 $A e^{it} \bar{v} = e^{-i \cdot 2t} \lambda (e^{it} v)$. 取 $\theta = -t/2$.

Exercise 3 Suppose (v, σ) is a eigenpair of A ; then $(v, \sigma\bar{\sigma})$ is an eigenpair of $A\bar{A}$. In particular,

- if $A \in \mathbb{C}^{d \times d}$ possesses d eigenpairs which are linearly independent over $\mathbb{C}$, then $A\bar{A}$ is diagonalisable with all eigenvalues non-negative.

Now consider the following **partial converse statement**:

- A has no eigenvectors if and only if $A\bar{A}$ admits no eigenvalues in $\mathbb{R}_{\geq 0}$;
- $A \in \mathbb{C}^{d \times d}$ has d linearly independent eigenpairs over $\mathbb{C}$ if and only if $A\bar{A}$ is diagonalisable with non-negative eigenvalues.

Prove the **partial converse statement**: suppose that $(v, \sigma\bar{\sigma})$ is an eigenpair of $A\bar{A}$; then σ is a eigenvalue of A .

错误解答: $u := \bar{\sigma}v + A\bar{v}$, 则 $A\bar{u} = \sigma A\bar{v} + \sigma\bar{\sigma}v = \sigma u$.

注: 应当要求 $u \neq 0$. 取 $u_t := e^{-it}\bar{\sigma}v + e^{it}A\bar{v}$, 则 $A\bar{u}_t = e^{it}\sigma A\bar{v} + e^{-it}\sigma\bar{\sigma}v = \sigma u_t$. 至少存在一个 t 使得 $u_t \neq 0$.

Exercise 4 Elucidate **Exercise 3** (particularly the warning) with the following example:

$$A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

由 $A\bar{A} = I$, v 是 $A\bar{A}$ 的特征向量, 但不是 A 的 eigenvector.

Exercise 5 This serves to complete **Exercise 2**. Suppose $\{(v_i, \lambda_i)\}_{i=1}^n$ are eigenpairs of A such that $|\lambda_i| \neq |\lambda_j|$; show that $\{v_i\}_{i=1}^n$ are linearly independent.

$\{v_i\}$ 是 $A\bar{A}$ 不同特征根的特征向量, 从而线性无关.

Exercise 6 We aim to identify the analogue of the Jordan canonical form in the c-version, namely, $X = PJ\bar{P}^{-1}$. We commence with a very special case:

- $P\bar{P} = I$ if and only if $P = Q\bar{Q}^{-1}$ for some invertible Q .

假定 $P\bar{P} = I$. 记 $Q_t := e^{-it}I + e^{it}P$, 此时 $P\bar{Q}_t = e^{it}P + e^{-it}I = Q_t$. 比较特征根, 存在 t 使得 $Q_t \neq 0$.

假定 $P = Q\bar{Q}^{-1}$, 则 $P\bar{P} = Q\bar{Q}^{-1}\bar{Q}Q^{-1} = I$.

Exercise 7 The most elementary Jordan matrix is the diagonal matrix. Show that A is cdiagonalisable (i.e., $A = P\bar{D}\bar{P}^{-1}$ for some P) if and only if $A\bar{A}$ is diagonalisable and $\text{rank}(A) = \text{rank}(A\bar{A})$. In this scenario, $A\bar{A}$ has only non-negative eigenvalues.

若 $A = P\bar{D}\bar{P}^{-1}$, 则 $A\bar{A} = P\bar{D}\bar{D}P^{-1}$ 是可对角化的矩阵.

若 $A\bar{A}$ 可对角化, 记 $P(A\bar{A})P^{-1} = D$, 其中 D 是以数量矩阵 $\lambda_i I_{s_i}$ 为分块的对角矩阵. 形式地取 $B := P\bar{A}\bar{P}^{-1}$, 以下计算表明 B 是分块对角矩阵:

- 由 $B\bar{B} = D = \bar{D} = \bar{B}B$, 得 $BD = B\bar{B}B = DB$. 由交换性, B 必然是分块对角矩阵.

此时, 对 A 的每一个分块作 c-对角化即可. 不妨设 $A\bar{A}$ 是零矩阵或者恒等矩阵.

- 若 $A\bar{A}$ 是零矩阵, 由秩关系得 $A = O$.
- 若 $A\bar{A} = I$, 则存在可逆矩阵 Q 使得 $A = QI\bar{Q}^{-1}$.

作为推论: 若 $\overline{A}A$ 可对角化, 则 A $\mathbb{C}$ -相似于某个全为非负实数的 Jordan 型, 从而 $\overline{A}A$ 的特征值均为非负实数.

Exercise 8 (Nilpotent part). Show that there exists a matrix P such that $PA\overline{P}^{-1} = \begin{pmatrix} N & O \\ O & B \end{pmatrix}$, where N comprises nilpotent Jordan blocks and B is invertible.

直接使用 Fitting 引理, 得两种直和分解: $V = \text{im}(\cdots \overline{A}A\overline{A}A) \oplus \ker(\cdots \overline{A}A\overline{A}A)$.

取 $\mathbb{R}$ -线性自同态 φ , 使得有 $\mathbb{C}$ -线性自同态 $\varphi \circ \overline{(\cdot)}$. 称这一自同态在基 u_i 下的 $\mathbb{C}$ -表示为矩阵 A , 若

$$\varphi(u_1 \mid \cdots \mid u_n) = (\overline{u_1} \mid \cdots \mid \overline{u_n}) \cdot A.$$

存在 $\mathbb{C}$ -相似变换 $PA\overline{P}^{-1} = A'$, 当且仅当存在一组基 $(v_i) = (u_i)P^{-1}$ 使得 $\varphi(v_\bullet) = (\overline{v_\bullet}) \cdot A'$. 比对 Fitting 引理,

- 存在 $\text{im}(\cdots \overline{A}A\overline{A}A)$ 的一组基, 使得在 $\varphi \circ \overline{(\cdot)}$ 下是同构;
- 存在 $\ker(\cdots \overline{A}A\overline{A}A)$ 的一组基, 使得在 $\varphi \circ \overline{(\cdot)}$ 下是幂零变换.

这就完成了 $\mathbb{C}$ -分块对角化.

考虑增长的零空间 $\ker A, \ker \overline{A}A, \ker A\overline{A}A, \ker \overline{A}A\overline{A}A$ 等等. 记 V_{-1} 是最后一个不为零的零空间, V_{-k-1} 是 V_{-k} 的前项. 这些都是互相包含的 $\mathbb{C}$ -线性空间, 因为 A 与 $\overline{A}$ 是向左添加的.

取 V_{-1} 的基 $\{v_{(0,\bullet)}^0\}$. 此时存在一组 $\{v_{(0,\bullet)}^1\}$ 使得 $Av_{(0,i)}^1 = v_{(0,i)}^0$. 另取 $\{v_{(1,\bullet)}^1\}$ 使得 $\{v_{(0,\bullet)}^1\} \sqcup \{v_{(1,\bullet)}^1\}$ 是 V_{-2} 的基. 归纳地, 取 $\{v_{(k,\bullet)}^2\}_{k=1,2,3}$ 使得有双射 $\overline{A} : \{v_{(k,\bullet)}^2\}_{k=1,2} \xrightarrow{\sim} \{v_{(k,\bullet)}^1\}_{k=1,2}$.

将归纳过程总结作下图 (同一般 Jordan 型的归纳):

$$\left[\begin{array}{cccccc} V_{-3} & \xrightarrow{\overline{A}} & V_{-2} & \xrightarrow{A} & V_{-1} & \xrightarrow{\overline{A}} & 0 \\ \{v_{(0,\bullet)}^2\} & \xrightarrow{\sim} & \{v_{(0,\bullet)}^1\} & \xrightarrow{\sim} & \{v_{(0,\bullet)}^0\} & \twoheadrightarrow & 0 \\ \{v_{(1,\bullet)}^2\} & \xrightarrow{\sim} & \{v_{(1,\bullet)}^1\} & \twoheadrightarrow & 0 & & \\ \{v_{(2,\bullet)}^2\} & \twoheadrightarrow & 0 & & & & \end{array} \right].$$

固定 (i, j) , 则一条链 $\{v_{(i,j)}^k\}$ 对应一个幂零 Jordan 块 $J(0)$. 矩阵 (视作线性映射) 的功效是平移链上的点.

Exercise 9 To analyse the Jordan form of A , we begin by studying the Jordan form of $A\bar{A}$. Demonstrate the following:

1. $A\bar{A}$ has a characteristic polynomial with real coefficients;
2. non-real Jordan blocks occur in conjugate pairs;
3. if $\dim \ker(\lambda I - A\bar{A}) = 1$ and $\lambda \in \mathbb{R}$, then $\lambda \geq 0$;
4. **(optional)** Generalise item 3 by showing that for any eigenvalue $\lambda < 0$, the dimension of the null space $\dim \ker(\lambda I - A\bar{A})^p$ is always even ($\forall p \geq 1$), implying that negative Jordan blocks occur in pairs.

(1.) 由方阵 $\det(xI - AB) = \det(xI - BA)$, 得 $A\bar{A}$ 与 $\bar{A}A$ 特征多项式相同. 结合 $\det(xI - A\bar{A}) = \det(xI - \bar{A}A)$, 该特征多项式是实系数的.

(2.) 讨论广义特征向量的对应关系即可. 若 $\lambda v = A\bar{A}u$, 则 $e^{it}\bar{\lambda}v = \bar{A}A(e^{it}u)$. 取 t 使得 $e^{it}\bar{\lambda} = \lambda$ 即可. 由于 t 是取定的, 故 $A\bar{A}$ 与 $\bar{A}A$ 关于 λ 的特征空间是同构的, 通过取共轭再数乘 e^{it} 实现.

(3.) 若 $\lambda < 0$, 取 $v \in \ker(\lambda I - A\bar{A})$, 则 $A\bar{A}(A\bar{v}) = \lambda(A\bar{v})$. 由特征空间一维, 必有 $zv = A\bar{v}$. 此时 $A\bar{A}v = z\bar{z}v$, 与 $\lambda < 0$ 矛盾.

(4.) 通过商空间归纳即可. 对 $\lambda < 0$, 假定 $\ker(\lambda I - A\bar{A})^k$ 是偶数维的, 往证 $\ker(\lambda I - A\bar{A})^{k+1}$ 也是偶数维的. 考虑限制在不变子空间 $\ker(\lambda I - A\bar{A})^k$ 上的线性变换, $\ker(\lambda I - A\bar{A})|_{\ker(\lambda I - A\bar{A})^k}$ 是偶数维的. 由 Rank-Nullity 与归纳假设得证.

Exercise 10 Show that there exists a matrix P such that PAP^{-1} is block diagonal with the following summands:

1. (Jordan type): $J_s(\lambda)$ where $\lambda \geq 0$;
2. (Half-Jordan type): Let $H_{2s}(\lambda) := \begin{pmatrix} O & I \\ J_s(\lambda) & O \end{pmatrix}$.
 1. (For non-real eigenvalues): Since each complex Jordan block in $A\bar{A}$ appears in a pair, i.e., $\begin{pmatrix} J_s(z) & O \\ O & J_s(\bar{z}) \end{pmatrix}$, define $H_{2s}(z)$ as a block of PAP^{-1} satisfying

$$H_{2s}(z) = \begin{pmatrix} O & I \\ J_s(z) & O \end{pmatrix}.$$

2. (For negative eigenvalues): For a negative eigenvalue λ of A , the Jordan blocks also appear in pairs (see **Exercise 9**).

We refer to this as the cJordan form of A . It is uniquely determined by A .

这是直接的验证. 关键点 (幂零部分的结构, 各阶广义负特征空间为偶数) 已在前文给出证明.

Exercise 11 (optional) (The uniqueness between the cJordan form of A and the Jordan form of $A\bar{A}$). The cJordan form is essentially unique.

- Find distinct matrices $A \neq B$ such that $A\bar{A} = B\bar{B}$;

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \text{ 与 } B = O.$$

- Suppose $A\bar{A} = B\bar{B}$ are invertible; then there exists a matrix P such that $PAP^{-1} = B$. In this case, we say A is scimilar to B .

在相差一个旋转 (e^{it}) 的情形下, $J(\neq 0)$ 与 $H(\neq 0)$ 部分唯一确定. 相差的旋转部分可通过 $e^{it/2} \cdot I$ (及其构成的分块对角矩阵) 的 c -相似所用抵消.

In summary, A is csimilar to B if and only if the following hold:

- $A\bar{A}$ is similar to $B\bar{B}$;
- $\text{rank}(A\bar{A}A \cdots A(\bar{A})) \equiv \text{rank}(B\bar{B}B \cdots B(\bar{B}))$; that is, A and B possess the same $J(0)$ -portion in their cJordan forms.

Exercise 12 (optional) Utilise the cJordan form to show that any complex matrix A is csimilar to:

- λA for $|\lambda| = 1$;
- $\bar{A}$;
- A^T ;
- a real matrix;
- a symmetric matrix

前三条就是直接的验证 ($A\bar{A}$ 相似, 且 J_0 部分相同).

对第四条, 直接构造分块实矩阵 $\begin{pmatrix} N & O \\ O & R \end{pmatrix}$ 使得 N 是 A 的 c -幂零部分, R 是 $A\bar{A}$ 可逆部分的平方根 (对复特征部分, 考虑 $\begin{pmatrix} 0 & + \\ - & 0 \end{pmatrix}$ 一类的旋转矩阵; 负特征部分总是成对出现, 从而存在实的平方根).

对第五条, PAP^{-1} 是对称的, 当且仅当 $PAP^{-1} = (PAP^{-1})^T$, 也就是 $(P^T P) \cdot A = A^T \cdot (P^T P)$. 由 Choleskey 分解, 复矩阵对称当且仅当其形如 $P^T P$. 由往期作业 (考察有理标准型以及多项式友矩阵), 任意域上的矩阵通过对称矩阵与其转置相似.

Exercise 13 (optional) Show that for any complex square matrix A ,

$$\det(AA^H + I) \geq \det(A\bar{A} + I) > 0.$$

由特征多项式实系数, 以上三项都在 $\mathbb{R}$ 中.

(第二个不等式). 由 $A = PR\bar{P}^{-1}$, 得

$$\det(A\bar{A} + I) = \det(R^2 + I) = |\det(I + iR)|^2 > 0.$$

(第一个不等式). 记增广矩阵 $T := (I \ A)$, 第一个不等式等价于 $\det T\bar{T}^T \geq \det TT^T$. 由 Cauchy-Binet 公式展开, 左侧是右侧的复数求和各项取模长所得, 取等当且仅当 A 是实矩阵.

Exercise 14 (optional) Using cJordan forms, show that any complex matrix is the product of a symmetric matrix and a positive definite Hermitian matrix.

由 Choleskey 分解与 Krull-Schmidt 酉正交化, 任意对称的复矩阵形如 $U^T D U$, 其中 D 是取值 $\mathbb{R}_{\geq 0}$ 的对角矩阵, U 是酉矩阵. 先将复矩阵 c-相似作对称矩阵, 得 PSP^{-1} . 将 S 写作 $U^T D U$, 得

1. 任意复矩阵是对称矩阵与正定 Hermitian 矩阵乘积, $PU^T D U P^T \cdot (P^T)^{-1} \bar{P}^{-1}$;
2. 任意复矩阵是可逆对称矩阵与半正定 Hermitian 矩阵乘积, $PU^T U P^T \cdot P^{T,-1} \bar{U}^T D U \bar{P}^{-1}$.

Exercise 15 (optional) Show that $\begin{pmatrix} O & A \\ A & O \end{pmatrix}$ is similar to $\begin{pmatrix} O & B \\ B & O \end{pmatrix}$, if and only if $\begin{pmatrix} O & A \\ -A & O \end{pmatrix}$ is similar to $\begin{pmatrix} O & B \\ -B & O \end{pmatrix}$.

若 $\begin{pmatrix} O & A \\ A & O \end{pmatrix}$ 与 $\begin{pmatrix} O & B \\ B & O \end{pmatrix}$ 相似, 则平方后亦然. 结合 $A\bar{A} \sim \bar{A}A$, 得 $A\bar{A} \sim B\bar{B}$. 比较零空间维数, 得 A 与 B c-相似.

若 A 与 B c-相似, 记作 $A = PB\bar{P}^{-1}$. 此时 $\begin{pmatrix} P & O \\ O & \bar{P} \end{pmatrix} \begin{pmatrix} O & A \\ A & O \end{pmatrix} \begin{pmatrix} P & O \\ O & \bar{P} \end{pmatrix}^{-1} = \begin{pmatrix} O & B \\ B & O \end{pmatrix}$.